Secure and Scalable Federated Learning for Distributed IoT Attack Detection

D.Boja Bindusri

1 Introduction

The Internet of Things has revolutionized countless industries by connecting ordinary things to the Internet, which allowed these ordinary things to share this abundance of data and automate many of these processes. The development of IoT devices, along with the implementation of 5G networks, has allowed for a reconstruction of modern communication infrastructures. This is due to the low latency of communications, a large number of devices able to be connected at the same time and the ubiquity of those connections in the new smart environments of industrial automation, connected healthcare and smart cities.

While the increased level of connectivity among IoT devices has brought about numerous opportunities for the development of new technologies and processes, there are also significant challenges associated with the security of these new devices and networks. One primary security concern involves managing the endless supply of vulnerabilities that arise as a result of the dependence on other IoT devices. In many cases this contributes to the overall risk to the data that these devices are intended to store or transmit i.e. data integrity, confidentiality and availability. Additionally, the widespread and heterogeneous nature of IoT networks results in the increased size of the attack surface. Many existing traditional security monitoring approaches will not function as intended when utilized in a distributed and heterogeneous operating environment.

There is a pressing need for scalable, secure, and privacy-preserving security mechanisms for IoT Networks. Federated learning advancements will be enhanced with Intelligent Model Architecture, as well as Adaptive Training Strategy, which in turn will enhance the federated system for use in IoT Security. New Adaptive Aggregation Technology will measure each participating device by assigning a weight based on the quality and reliability of any data or contributions they make to the system and improve the federated model as a whole. The continuing advancements occurring in Federated Learning will lead to the development of future solutions that are increasingly secure, efficient and scalable for use in detecting current Intrusions in an Existent IoT ecosystem[9, 12, 14].

1.1 Security Vulnerabilites in IoT Environments

IoT devices are typically designed with minimal levels of security, often leading to devices being exposed to a variety of threats. Procedures for authenticating users utilizing weak authentication measures provide unauthorized access to the IoT system and facilitate manipulation of the devices. Data transmitted between IoT devices and their respective servers is, due in part to the lack of appropriate encryption methods, exposed to unauthorized Users' access to the system. Lack of regular software updates to the IoT devices has also left a significant number of devices susceptible to previously identified security vulnerabilities. These security vulnerabilities reflect a critical need for scalable and effective security measures that can operate in environments where there are limited computing resources [7].

A botnet consists of compromised devices – frequently IoT devices – that an attacker uses to complete malicious tasks. These botnets take advantage of weaknesses in connected devices and turn them into hacked nodes that are controlled through a command-and-control server. Botnets perform several different types of attacks, including Distributed Denial-of-Service attacks, the theft of data, credential stuffing, and ransomware. Botnet attacks have also grown increasingly sophisticated and more difficult for network security professionals and organizations to detect due to the rapid growth of the IoT ecosystem [8]. Botnet attacks pose an enormous risk to the integrity of critical infrastructure such as transportation systems, financial services, healthcare, and the privacy of user data.

Smart home devices such as cameras, sensors, and connected appliances have made our lives easier through increased efficiency and automation. However, the wide variety of smart home device types, smart home configurations, and how each user interacts with their smart home makes it incredibly challenging to create highly effective intrusion detection systems (IDSs) for smart home environments. Smart home networks can be as simple as 1-2 smart devices or as sophisticated as multiple smart home networks connected to a variety of different automation systems to the entire building. Add to this the fact that devices are constantly joining/leaving the network, and thus maintaining the security of smart home devices will become increasingly difficult. This will also increase the attack surface for botnets and demonstrates the need for improved and adaptive intrusion detection systems[17, 17].

1.2 Intrusion Detection in IoT Networks

Intrusion Detection Systems (IDS) play a critical role in analysing network traffic to detect unauthorised activities. Among them, anomaly-based IDS are particularly effective, as they identify deviations from expected behaviour by comparing network activity against predefined patterns. Leveraging machine learning, these systems enable real-time threat

detection and rapid response [9]. Machine learning improves the detection capacity of recognizing malicious traffic in network environments.

IoT devices generate large volumes of data continuously, making real-time anomaly detection a crucial requirement for securing IoT networks. Traditional anomaly detection models rely on batch processing that requires full dataset analysis. However, this approach is computationally infeasible for real-time IoT networks due to memory and processing limitations. As a result, new detection mechanisms must be designed to handle high-speed streaming data efficiently [15].

IDS continuously monitor network traffic and device activity, enabling early detection of potential threats and serving as a critical layer of defense. Traditional IDS approaches face significant limitations in IoT settings due to reliance on centralized cloud systems for data analysis. These centralized solutions increase the risk of data breaches and impose high bandwidth and energy costs for resource-constrained IoT devices. Machine learning models rely on sufficient data samples to train accurate detection models . However, many IoT nodes are application-specific and may not generate enough training samples for effective machine learning-based detection. The threat landscape in IoT environments is highly dynamic due to the rapid growth of devices and applications . Static models may fail to cope with rapid changes in attack patterns and data distributions [10].

1.3 Limitations of Centralized Security Approaches

Traditional intrusion detection systems are typically based on using a centralized system to process information. Therefore, centralizing intrusion detection creates significant privacy and communication issues. Additionally, centralizing intrusion detection does not meet the performance requirements of IoT systems, nor does it have the latency or scalability necessary to support the high volume and speed of data generated by IoT devices. As such, the amount and speed at which data from IoT devices are generated exceed the capabilities of traditional IDS and traditional IDS tend to fail to meet real-time processing window. In addition, centralizing the collection of data increases the opportunities for the risk of disclosure of sensitive data and violation of privacy[14].

The traditional methods for botnet detection also utilize centralized data aggregation and analysis methods and as such they have challenges with scalability and substantial computational overhead. Therefore, botnet detection methods are not well suited for IoT environments, as the dynamic and distributed nature of IoT environments frequently change in terms of network conditions and attack patterns. Thus, the organizations using traditional models to detect botnets in IoT environments could experience decreased precision, particularly in Industrial IoT (IIoT) environments.[**?**].

Centralized systems also require a constant flow of data in high quantities to a specific server, which leads to problematic performance levels due to increased network traffic,

increased latency, and increased energy consumption for any end-user connected to that central site; it therefore creates a single point of failure on the entire network, allowing for targeted attacks to take place against the entire system. As the growth of the Internet of Things (IoT) grows exponentially, centralized monitoring will become increasingly difficult and inefficient to perform.

1.4 Federated Learning for Distributed IoT Attack Detection

Federated Learning (FL) is a new method of using distributed machine learning to perform collaborative model training across devices without having to transfer any raw, local data between the devices. Using FL for aggregating and improving model performance provides increased privacy, since the data remains local and is used only to create and contribute to an improved global model. Thus, FL eliminates the risk of data leakage or privacy violations while also providing the ability to perform distributed learning very efficiently.

Within a FL environment, the participant will download a global model from the server, train that global model locally with its private data, and send only the update to the server (not the private data). The server can aggregate the model updates, then improve the global model; this updated global model will be distributed back out to all participants, allowing them to share in any improvements as a result of the collectively established learning among the entire group.[14].

In addition to the fact that FL removes the need to transfer significant amounts of data across a network, FL also provides much greater scalability benefits, etc. Federated learning has lower latency and higher flexibility than traditional centralised machine-learning frameworks in addition to being well-suited for quickly-changing IoT settings; by allowing for high-performance, yet scalable, distributed IDSs to be created in an IoT environment, it is an excellent candidate for implementing secure, scalable intrusion detection systems within a distributed IoT environment. IoT and Distributed IoT Architectures.

2 Background and Conceptual Foundations

2.1 Internet of Things (IoT) and Distributed IoT Architectures

The Internet of Things (IoT) is an evolving technology that connects everyday objects together through Internet connectivity, enabling the sharing of information and data, automation of activities, and an ability to monitor something in real time. In an IoT environment, components such as sensors, actuators, embedded devices, and others communicate via networks in order to collect and process information from the environment around them, thus producing intelligent decisions that will be utilized throughout vari-

ous sectors in the world. IoT has many uses, including smart cities, health care, transportation, industrial automation, and environmental monitoring; all of which depend on efficient real-time data processing. Typically, IoT architectures consist of four basic components; they include sensing devices, communications networks, edge nodes, and cloud platforms. Sensing devices collect information about the real world and transmit it to a gateway via a communication network for further processing and analysis. While cloud platforms can manage huge amounts of data storage and computing resources on a large scale, the resources located at the edge can process data closer to where it is captured, thereby decreasing latency and improving overall performance of the system [1, 8].

Distributed architectures function via systems of connected devices that are located far apart, using a variety of communication technologies (e.g., Wi-Fi, Bluetooth, ZigBee, cellular). They can also facilitate scaling up IoT systems and make it easier to move data from many locations to other locations. Increased interconnectivity of devices has resulted in increased complexity of IoT systems. Most IoT devices have limited resources (e.g., computational power and energy), making it difficult to implement superior processing and security solutions. Managing communication, processing and storage of data among dispersed systems poses a major challenge. Accordingly, to support continued growth in the IoT infrastructure, effective use of resources, reliable communication, and systems designed for scalability are critical[13, 7].

2.2 Security Challenges and Cyberattacks in IoT Networks

With the rapid growth of IoT technologies, the risk associated with cyber threats has dramatically increased (e.g., the large number of IoT devices operating with low computational power, limited memory, and low energy). Therefore, IoT devices typically lack significant security mechanisms such as strong authentication, encryption, and regular software updates, which means that IoT devices are more likely to be attacked or compromised by unauthorised parties using malware or attacking the underlying network. Consequently, securing IoT across distributed systems is challenging and allows malicious actors to exploit security vulnerabilities to perform attacks or disrupt operations within the network [7, 1].

Botnet attacks present one of the primary dangers faced by Internet-of-Things networks. Botnets consist of hijacked devices (also known as "zombie" devices) that can be controlled remotely by attackers to carry out harmful behaviour. The large number of compromised devices makes it easy for botnets to launch attacks across distributed networks. Distributed denial-of-service (DDoS) attacks are another large threat because they rely on compromised devices to create large amounts of traffic that can overburden a target device. In an Internet-of-Things environment, botnet and DDoS attacks can scale quickly due to the high number of connected devices and significantly affect

performance[15, 7].

Cyberattacks are getting more sophisticated and frequent, which means that we need to develop more advanced security mechanisms for real-time applications. Most of the traditional security methods have been found ineffective because the methods are based on a centralised architecture, and are unable to efficiently support the large-number of distributed data. Therefore, we are in need of intelligent intrusion detection techniques that can detect abnormal behaviours in the constantly changing Internet-of-Things landscape[8].

Table 1: Literature Survey of Existing Approaches for IoT Attack Detection

Ref	**Method Proposed**	**Techniques Used**	**Issues Resolved**	**Limitation**	**Performance**
	FL-based DDoS detection for SDN	FL, Distributed Training, SDN	Improves DDoS detection with privacy preservation	Aggregation communication overhead	$\approx$ 98% accuracy
R13	Property testing-based IoT anomaly detection	Property Testing, Edge AI, FL	Reduces computational cost in IoT IDS	Depends on probabilistic sampling	$\approx$ 92% accuracy
R14	Privacy-preserving FL for botnet detection	FL, SHAP, Distributed Training	Detects botnet attacks with data privacy	Non-IID data aggregation issues	$\approx$ 99.98% recall
R15	FL-based intrusion detection system	FedAvg, Machine Learning	Distributed IDS with privacy preservation	Affected by heterogeneous data	$\approx$ 85–89% accuracy
R16	Edge-FLGuard real-time detection framework	FL, Edge AI, LSTM, Autoencoder	Real-time scalable detection in 5G-IoT	Vulnerable to poisoning attacks	F1-score $\geq$ 0.91
R17	Hybrid ensemble FL intrusion detection	Ensemble Learning, KNN, RF, FL	Improves accuracy, reduces false positives	Weak for rare attacks	High TPR, low FPR
R18	Blockchain-enabled FL intrusion detection	FL, Blockchain, Knowledge Distillation	Enhances trust and secure collaboration	High computational complexity	$\approx$ 98.6% accuracy

As shown in Table 1, intrusion detection systems that use federated learning allow for high levels of detection accuracy while maintaining the confidentiality of the data being processed[4, 9]. Hybrid deep learning techniques have shown promise in identifying complex types of attacks, and lightweight models tend to minimise the amount of computational power required by resource-limited devices. However, communication overhead and heterogeneous data distribution pose additional challenges that suggest we need adaptive and secure frameworks in which to conduct our work.

2.3 Intrusion Detection Systems and Machine Learning for IoT Security

Intrusion Detection Systems (IDS) play an important role in protecting computer networks and IoT infrastructures from malicious activities. IDS captures both inbound and outbound network traffic and watches how users behave on a system in order to catch anything that looks like it doesn't belong there, as well as identify any abnormal activity

that might mean a cyber attack is in progress. In IoT environments, this is particularly important due to the high number of devices connected, and the sheer complexity of a distributed network. There are two basic types of intrusion detection techniques: signature-based and anomaly-based methods. Signature-based methods examine known attack patterns to identify them as they come in; therefore they are very useful for identifying attacks that have previously been identified. They are not able to detect new or unknown types of attacks. On the other hand, anomaly-based methods develop a baseline of standard behaviour of the network, and any deviation from that baseline is potentially an attack. While the ability of anomaly based methods to detect unknown types of attacks is strong; they can produce higher false positive rates than signature based intrusion detection systems[1, 9].

Machine Learning's role in making Intrusion Detection Systems (IDS) better at detecting intrusions through accuracy and efficiency increase has seen significant growth in the area of IDS, with most machine learning algorithms working directly on the basis of how they analyze network traffic data (ND) — learn how to distinguish between normal and malicious patterns of network traffic data and discover complex representations of the relationship between the two data sets. For this reason, Machine Learning has become one of the most important components of today's IoT systems security architecture[1]. Typical IDS that utilize a centralized architecture must continuously collect data on their network environment , process the collected data at some central location(s) and have the network send all collected data to their central server for processing and analysis. This results in an increase in the network's overall communication overhead and latency, as well as an increase in privacy concerns because transmitting the collected data from all or even a significant number of devices to a single, central server increases the chance of the data being leaked [9, 8].

As IoT devices continue to grow at an exponential rate, there is a desire for scalable and efficient security solutions. The current networked intrusion detection solutions must be capable of processing large amounts of data (from many devices), maintaining the privacy of the individual device data, and reducing or eliminating the communication overhead associated with working with collected device data.

2.4 Federated Learning for Privacy-Preserving Distributed Attack Detection

Federated Learning (FL) is an innovative distributed machine learning (ML) model that allows numerous devices to create a global model collaboratively while keeping their original form of data private. Devices train a local model on their own respective datasets and send the model updates only to an aggregation server (the server that all participating devices send their respective model updates). The server will then combine the model

updates sent by the different devices and create a single, better version of the global model, returning it to all participating devices. FL is a new way to conduct machine learning while preserving the confidential nature of the original data, thus allowing for collaborative model learning across different device types in different locations[16].

When you use traditional centralized machine learning approaches, you would need to send all the data to the central server for processing. This has a high cost regarding communication overhead, increasing communication latency, and the associated privacy problems. However, with FL, the data remains on the devices, and the parameters that comprise the models are shared, thus reducing the amount of time that data must be transmitted throughout the system[1].

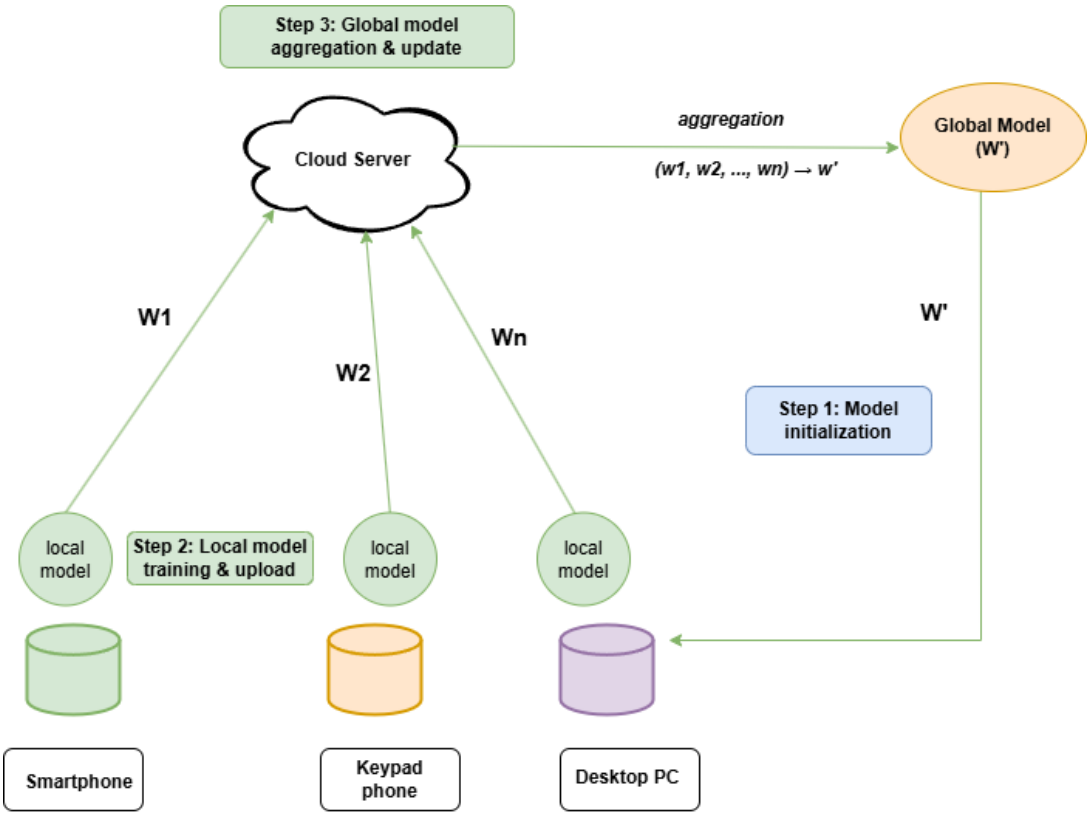

Figure 1: Federated Learning Workflow for Distributed IoT Attack Detection

The above Figure 1 illustrates the workflow of a federated learning-based intrusion detection system, where IoT devices perform local training and share only model updates with a central server. The server aggregates these updates to generate a global model, which is redistributed to the devices, enabling collaborative and privacy-preserving attack detectionn[3].

FL systems allow the participating devices to train their models using local data that they do not share with anyone other than the central server through model updates. The central server aggregates the model updates and provides an improved version of the global model, which is then redistributed back out to all of the participating devices.

The entire process is repeated for multiple rounds until the model is satisfactory, allowing for privacy-preserving scalable learning. However, challenges such as heterogeneous data distribution, communication overhead, and security threats like model poisoning attacks still exist. Addressing these issues is essential for developing secure and scalable federated learning-based intrusion detection systems.

3 Core Concepts and Approaches

The Internet of Things (IoT) refers to a network of interconnected devices that communicate through the internet to collect, exchange, and process data. IoT devices include sensors, actuators, embedded systems, and communication modules that interact with each other to perform automated tasks. These devices enable real-time monitoring and intelligent decision-making in various applications such as healthcare systems, smart homes, industrial automation, and transportation networks. The rapid expansion of IoT technologies has significantly increased the number of connected devices, creating large-scale distributed environments that continuously generate and exchange data. IoT systems typically follow a layered architecture consisting of sensing devices, communication networks, edge nodes, and cloud platforms. At the sensing layer, devices collect environmental information such as temperature, motion, or network activity. This data is transmitted through communication networks to gateways or edge devices, which manage connectivity and perform preliminary processing tasks. Cloud platforms are then used to store and analyze large volumes of collected data, enabling advanced analytics and intelligent services[7].

Centralized processing methods to compute this type of large data set increase time delays of transmitting data, as well as, higher bandwidth usage that may occur from transferring the aforementioned quantity. Therefore, with regards to improving systems scalability and performance, distributed computing also called cloud-computing or cooperative-computing methods are quickly being applied to the IoT environment.

In addition to being the main method for addressing these issues (scalability and performance), edge computing has become a requirement within the design of present-day IoT systems by enabling computing capability to exist at, or near, where the data was created. By executing the initial phase of data analysis at the edges of networks, latency and response time associated with real-time applications can be lowered, as well as, decreasing the amount of data transmitted to central servers resulting in greater network efficiency. Accordingly, because edge computing greatly aids in managing the high volume of distributed IoT infrastructure; its presence is particularly essential in overcoming challenges that all IoT devices face such as: limited available computing power; limited energy; lack of ability to use sophisticated processing and security algorithms due to low device computing resources and energy from devices. [7]. As IoT networks continue to

expand, efficient data management and secure communication across distributed devices become critical requirements for maintaining reliable system operations [13].

3.1 Intrusion Detection Systems for IoT Security

Intrusion Detection Systems (IDS) are essential security mechanisms used to monitor network activities and identify malicious behavior in IoT environments. IDS analyze network traffic and system events to detect unauthorized access, suspicious activities, and potential cyberattacks. Due to the large number of interconnected devices and the distributed nature of IoT systems, IDS play a critical role in maintaining network security and ensuring reliable system operation[1, 9].

There are two primary types of intrusion detection systems (IDS) that exist today: signature-based IDS detectors and anomaly-based IDS detectors. With signature-based IDS detectors, the detection of potential intrusions takes place when the IDS has predefined patterns of previously detected attack signatures. As a result, signature-based IDS are great for detection of previously detected intrusions; however, they are not as efficient for detection of new and previously undetected intrusions. Conversely, anomaly-based IDS detectors are, in general, based on a baseline of normal network behaviour and will identify potential intrusions as any deviations from this baseline. Thus, anomaly-based IDS systems may be used for detection of unknown intrusions, although they typically result in a higher number of false positives[8].

Machine learning-based algorithms have greatly improved intrusion detection by allowing the automatic analysis of large volumes of network data and detection of complex attack patterns. Additionally, machine learning algorithms have the ability to learn patterns from past data and become more accurate as more data is provided to them for training. Thus, machine learning-based IDS systems are now a critical part of the IoT security framework. However, a significant disadvantage of traditional IDS solutions is that they typically employ a centralised architecture where data from multiple devices must be sent to a single server for analysis. As a result, traditional IDS solutions create additional communication overhead, increased latency, and privacy concerns. Furthermore, many of the advanced deep learning models, such as the (convolutional neural networks) CNN and long short-term memory (LSTM) models, present a challenge to implement in an IoT system due to limited computation and energy resources. Therefore, researchers are focused on finding efficient, distributed, and scalable intrusion detection systems.

3.2 Federated Learning for Distributed Security

Federated Learning (FL) is a distributed machine learning approach that enables multiple devices to collaboratively train a global model without sharing raw data. In this approach, each device trains a local model using its own dataset and sends only model updates to a

central server. The server aggregates these updates to generate an improved global model, which is then redistributed to participating devices. This iterative process continues until the model achieves the desired performance, enabling collaborative and privacy-preserving learning [16]

Federated learning provides an excellent solution for IoT environments, where there is a large amount of continuously generated data from many different and geographically distributed devices. In contrast to traditional centralized models which require a large volume of data to be sent to and processed by a single central server with significant communication overhead, high latency due to transmission required, and privacy concerns, FL allows data to remain local to the device and permits learning via the sharing of parameters reducing the amount of data that needs to be transmitted and increasing overall system efficiency. Therefore, FL is highly effective at performing both intrusion detection and anomaly detection in an IoT network. Despite the numerous advantages of federated learning, there are also several challenges to overcome. One of the primary concerns is the communication overhead required to aggregate and synchronize the model across devices, particularly in the case of IoT networks that contain a large number of devices. Additionally, FL systems are also vulnerable to attacks from various types of models that have been poisoned, adversarial examples, etc. While researchers have developed various adaptive aggregation and hybrid deep learning methods that increase the robustness of federated learning systems and improve detection rates, there remains a significant opportunity for researchers to continue to work on creating safe, secure, efficient, and scalable FL frameworks. However, developing secure, efficient, and scalable federated learning frameworks remains an important research challenge [8, 14].

3.3 Secure and Scalable Attack Detection Frameworks

In recent years, numerous advances have been made concerning new frameworks that use distributed computing, as well as machine learning and federated learning techniques to help make IoT systems secure. These types of frameworks focus on providing efficient, scalable, and privacy-preserving ways of identifying and reducing risks associated with cyber security threats for systems that are distributed across multiple Internet-connected devices (IoT). When a device is connected to another through the edge of the network and the two collaborate to identify and respond to threats, this reduces the dependency on central systems for threat detection and response by providing a mechanism for local processing of device-generated data [13, 8].

The role of edge computing in these frameworks cannot be overstated. By enabling real-time processing of data as close to the source of the data as possible, edge computing can provide a significant reduction in latency, response time, and overall increase in efficiency to the system. Furthermore, when used in combination with federated learn-

ing methods, edge computing allows multiple devices to collaboratively train models while ensuring the confidentiality of their data. Utilisation of blockchain technology for improved security, authentication, communication and model update processes through decentralisation.

While the potential capability of these frameworks is enormous, several outstanding issues must be addressed regarding balancing detection accuracy, computational efficiency and system security concerns. IoT resource constraints of devices limit the utility of deploying complex algorithms, while communication resource constraints (overheads) limit the scalability of these frameworks. Future research shall focus on designing lightweight algorithms, developing more efficient communication methodologies, and providing secure methods for aggregation to enable the practical implementation of IoT solutions at scale. The fusion of distributed learning, edge intelligence, and collaborative security techniques is a viable opportunity for the development of next-generation intrusion detection systems design[8, 14].

4 Comparative Discussion

There are many techniques available for detecting and mitigating attacks on Internet of Things (IoT) networks. The major goals include enhancing accuracy of detection, maximizing scalability and increasing privacy of the data collected by the IoT system. Generally, most methods utilize machine learning, anomaly detection, and/or federated learning algorithms to identify malicious behavior in the IoT network traffic. However, due to the significant amount of data produced within IoT environments, the high heterogeneity of such devices, and insufficient computational capacity, no single technique can adequately address the security needs of the entirety of an IoT system's operation.

One type of detection approach that may provide accurate results is that of a centralized or cloud-based detection system, however, since all data is analyzed from one single source, there will be a higher amount communication overhead and possible latency problems and privacy issues can present with utilizing a centralized detection system. In contrast, using distributed computing or edge computing, allows all device data to process independently on a local level and will typically result in much lower latency than a centralized detection approach. Therefore, distributed computing and edge computing also provide a higher level of scalability compared to centralized detection. In addition, federated learning allows for the training of multiple collaborative detection models without having to share data, and hence, encouraging better privacy of the data, while still obtaining an adequate level of detection capability.

There are currently still some barriers that exist within current IoT system being able to detect cyber-attacks, with several advancements toward detection currently available, communication overhead associated with model update sharing, heterogeneous data

distribution across each device, and being subject to the aggressiveness of potential cyberattackers. These barriers highlight the continued need for more efficiency, security, and adaptability of attacks and their detection within IoT environments[16, 14].

4.1 Centralized Intrusion Detection vs Federated Learning Based Detection

Centralized Intrusion Detection Systems (IDSs) use a centralized server to collect and analyze network traffic data that has been aggregated from many Internet of Things (IoT) devices, allowing for improved accuracy. However, there are also drawbacks to this methodology. The need to transmit large volumes of data has resulted in an increase in communications overhead, consumption of significant bandwidth, and possible exposure of confidential or sensitive information, which adds additional privacy concerns. As a result of the continued expansion of connected devices within large-scale IoT ecosystems, centralized IDSs also suffer from latency and scalability issues[1].

Conversely, the use of Federated Learning-based IDS frameworks provides an alternative by allowing for the collaborative development and training of models using IoT devices, while also allowing for data privacy and retention of the integrity of the data produced by each device. Instead of transmitting large amounts of information to a centralized server, when using federated learning, each IoT device trains its own model locally and only transmits the model updates to the central aggregation server. By significantly reducing the amount of data transmitted over the network between the IoT devices and the central aggregation server, as well as by allowing the use of data to remain solely on the local device, federated learning represents an excellent option for the deployment of IDS in a large-scale IoT environment[16].

While federated learning represents a significant advancement over traditional centralized methods used for IDS, there are still a number of challenges associated with using this type of methodology. Federated learning systems are also susceptible to security concerns like model poisoning attacks perpetrated by compromised devices (i.e., devices that have been tampered with). Addressing these issues is critical to developing a secure and efficient federated learning framework for detecting attacks within an Internet of Things (IoT) system[8]. In general, centralized systems have excellent capabilities for detecting certain types of events; however, a federated learning framework implementation has a higher degree of scalability and a lower dimension of privacy loss when implemented in a modern IoT environment.

4.2 Machine Learning Based Detection vs Edge Intelligence Approaches

Machine learning-based intrusion detection systems utilize machine learning techniques to analyze network traffic for identifying patterns associated with malicious activities. Using two different types of techniques, IDSs are able to learn what constitutes normal and abnormal behaviour based on their ability to detect threats within an IoT system [9]. IDSs have demonstrated positive results by identifying complex data patterns and thereby improving their ability to detect future threats. However, traditional IDSs have primarily used a centralized processing model, which results in increased latency and communication overhead when performing data analysis on large volumes of data produced by IoT devices typically [9].

Edge Intelligence enhances Machine Learning by allowing data processing to be done on IoT devices instead of on centralized servers for lower latencies and faster reaction times to the detection of security threats. Also reduces the amount of time spent transmitting data since less data will need to be moved around. Reduces communication costs, improving overall efficiencies, when combined with distributed learning, and Edge Intelligence enables large-scaled intrusion detection capabilities across the Internet of Things (IoT) networks[8].

Edge Intelligence has some limitations due to a lack of computational power, memory and a limited energy supply of IoT devices. Because of these factors, complex ML Models are not able to run on IoT devices, and a separate set of lightweight and optimized algorithms must be developed instead. In addition, coordination across distributed Edge Nodes can be very difficult[13]. The combination of the strengths of Machine Learning, which provide excellent intrusion detection capabilities, with the enhancements offered through Edge Intelligence, such as improved real-time performance and scalability, will enhance efficient use of intrusion detection capabilities in an IoT environment, although there are resource constraints requiring careful optimization. As summarized in Table 2, centralized approaches provide high detection accuracy but suffer from privacy and scalability issues. Federated learning offers a balance between privacy and performance, while edge-based approaches enable real-time detection with reduced latency. Blockchain-integrated frameworks further enhance trust and security, although they introduce additional computational overhead.

4.3 Privacy-Preserving Detection vs Blockchain Supported Security Frameworks

Privacy Preservation is an important principle in IoT Security as any network traffic over the IoT contains some type of private information or sensitive information about the

Table 2: Comparison of Existing IoT Attack Detection Approaches

Approach	**Key Characteristics**	**Advantages**	**Limitations**	**Ref**
Centralized IDS	Central server analyzes network traffic	High accuracy with large datasets	High communication cost and privacy risk	R15
Federated Learning IDS	Local training with global aggregation	Privacy preservation and scalability	Non-IID data and communication overhead	R11, R16
Machine Learning IDS	Uses ML models for anomaly detection	Detects known and unknown attacks	Needs large data and computation	R15, R17
Edge-based Detection	Detection at edge devices near source	Low latency and real-time detection	Limited device resources	R16
Blockchain + FL Security	FL integrated with blockchain framework	High trust and secure collaboration	High system complexity and cost	R18

device or user. An important method of maintaining privacy is the recent trend to utilize Federated Learning, which allows for the training of machine learning models without moving the raw data to a centralized server. With Federated Learning, the data remains on the local device therefore reducing the potential of a data leak from the device and providing greater assurance of user privacy[1].

Certain other approaches to improve trust and security capitalize on integrating blockchain with federated learning to increase both separately. Specifically, blockchain acts as a decentralized record-keeping system of model updates while also verifying that only devices participating in the learning process are those that have been approved. This ultimately creates an increase in the amount of transparency, integrity and reliability that exists in collaboratively established intrusion detection systems that are operating in wide area distributed networks[14].

However, the addition of blockchain to federated learning incurs a significant amount of additional processing overhead due to both the additional processing resources being required for carrying out the blockchain operations themselves, and an increase in latency being incurred when implementing large scale IoT deployments. For this reason, the design of a framework that efficiently balances privacy, security, and computational efficiency is particularly difficult. Ultimately the direction for future work should lie in optimizing blockchain systems integrated with federated learning systems such that the amount of processing overhead is minimized while providing as strong a security guarantee as possible[8].

5 Practical Insights and Use Cases

When deploying devices in a practical IoT environment, they are typically restricted to limited resources and are vulnerable to various forms of attack, so the need for effective and scalable forms of intrusion detection becomes critical. Traditional methods of centralized intrusion detection are problematic due to their high level of communication overhead and the privacy issues they raise regarding the use of personal information. These two factors together make them impractical in today's large scale environments. In order to address these issues, federated learning allows for training models in multiple disparate locations, without sending raw data to a central location. Each individual device uses its own data to produce a local model and transmits only the local model's updates back to the central site, thus maintaining individual privacy and allowing for collaborative model training. This ability to conduct model training in this manner can also take advantage of edge computing technologies to allow for timely monitoring and faster detection of anomalous network activity. Essentially, by using edge devices to analyze traffic locally and contribute to a global model, organizations will improve detection accuracy and scalability[16, 13] In conclusion, the combination of federated learning and edge computing provides significant benefits to the implementation of an intrusion detection system by reducing communication costs yet maintaining performance in large scale IoT applications.

5.1 Smart Home IoT Security

IoT devices that are able to work together in the form of smart homes include: cameras, intelligent thermostats, smart locks and voice activation systems. These devices constantly share data with one another through home networks in order to create automation and provide intelligent control. Unfortunately, The increased number of IoT devices also creates an extensive attack surface for cyber threats. Attackers can take advantage of IoT devices vulnerabilities (bugs) to gain unauthorised access into those device networks or use those devices as infects that allow them to create botnets (groups) of infect devices that can be controlled by attackers[15].

Intrusion detection systems placed in smart homes are able to monitor traffic throughout the devices and detect atypical communications to identify what is malicious activity. Machine learning can be used to create an analysis of the device network behaviour and provide insight in possible deviations from normal traffic. Additionally, Federated Learning frameworks allow smart devices to work collaboratively to improve intrusion and attack detection models without transmitting actual users' sensitive data. In this case, the Federated Learning solution improves accuracy while also ensuring user privacy is protected from the use of central servers storing sensitive user's data.

5.2 Smart Systems

Smart systems such as smart cities, healthcare, and industrial environments rely on Internet of Things (IoT) devices for automation, monitoring, and intelligent decision-making. These types of infrastructures constantly generate and transfer data through multiple interconnected and distributed devices.The contribution of Edge Computing to the smart system is significant through the ability to locally process data, ultimately leading to reduced latency in supporting real-time decisions and efficiently supporting the operation of the system [11]. The rapid increase in the number of IoT devices has created a much larger potential point of attack for cyber threats, resulting in the need for scalable, real-time security solutions. A majority of IoT devices are deployed in environments where resources (i.e., computational power, memory, and/or energy) are limited; therefore, they cannot support the deployment of traditional security mechanisms [4, 11].

Intrusion Detection (IDS) has become a popular means of monitoring network activity, particularly for identifying Distributed Denial of Service (DDoS), Probe, Remote to Local (R2L), and User to Root (U2R) attacks; many of which can resemble legitimate traffic and be difficult to identify. The use of centralized IDS methods can lead to large communication overhead, latency, and violation of privacy due to the transfer of large quantities of sensitive data.

Federated learning can provide a decentralized method through which devices can collaboratively create a global model without having to share the raw data with other devices. This method will reduce communications overhead and maintain data privacy while allowing scalable methods of detecting potential intrusions across distributed systems[6].

Figure 2 illustrates a practical architecture for secure IoT environments, where edge devices perform local intrusion detection, federated learning enables collaborative model training, and blockchain ensures secure aggregation and trust management.

Smart environments' degree of heterogeneity has further increased the complexity of detecting attacks. For example, in smart homes there are many different kinds of devices that have numerous ways of being configured and used, making it difficult to develop uniform detection mechanisms [2]. Similarly, in industrial IoT environments, fluctuating network conditions pose challenges for developing robust detection systems[2].

Although machine learning techniques help improve detection accuracy, implementing complex models on resource-constrained devices remains a challenge. Therefore, federated learning integrated with edge computing enables scalable and privacy-preserving security solutions for modern smart systems.

5.3 Distributed IoT Attack Detection Systems

Distributed networked IoT consists of a large number of disparate devices operating in a variety of locations. Detecting cyber-attacks within such an environment will require

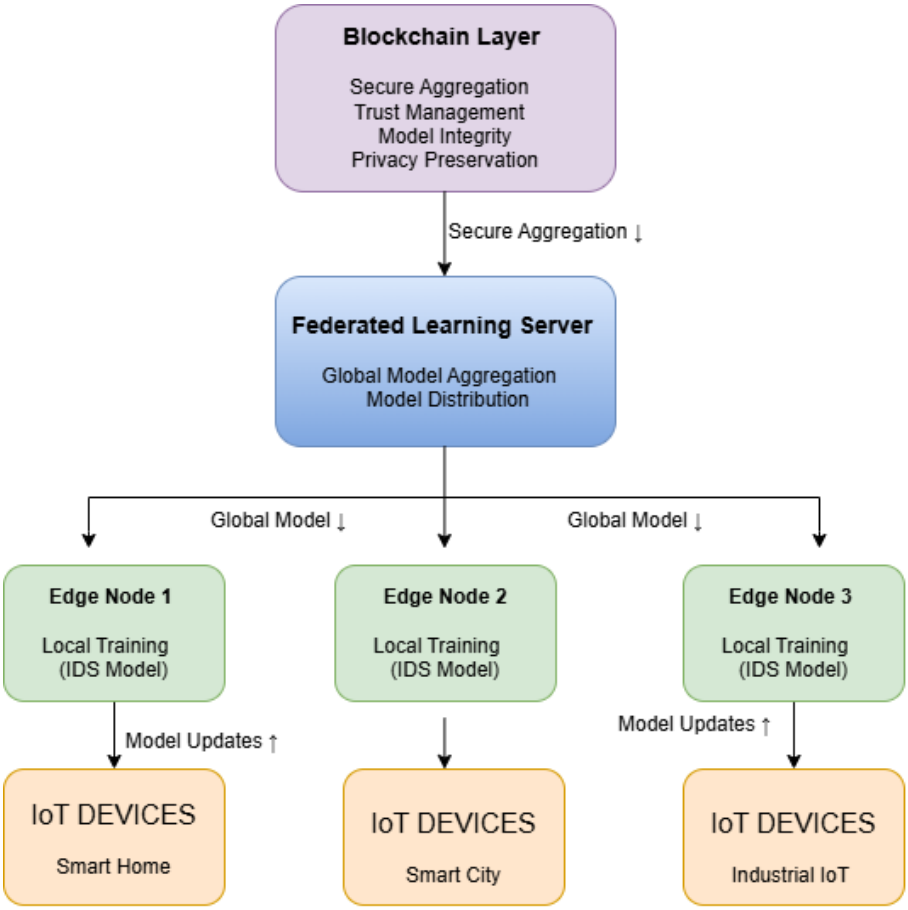

Figure 2: Integrated Edge, Federated Learning, and Blockchain-based Framework for Secure IoT Attack Detection

a scalable security mechanism capable of performing analysis against distributed data sources. Machine-learning-based intrusion detection systems can detect and identify network behaviour patterns associated with malicious activity and offer anomaly detection in real-time[1].

Federated learning frameworks represent a collaborative approach to distributed attack detection within an IoT environment. Each device participating in the detection system will contribute to the training of a global detection model while maintaining data privacy for the local device, thereby developing an iterative basis for scaling intrusion detection systems that are able to adapt to changing forms of cyber threats across distributed IoT infrastructures.[16].

6 Challenges and Open Issues

6.1 Resource Constraints

Many IoT nodes cannot execute advanced training tasks required by machine learning-based intrusion detection systems due to limited processing, memory, and power capabilities. IoT devices generate continuous streams of data in distributed IoT environments, necessitating real-time data analysis. However, low processing capabilities make it difficult to perform large-scale data analysis. Wireless sensors deployed in remote locations can only operate with battery-operated devices, which limits the development and deployment of advanced anomaly detection frameworks and models in the cloud[3].

Therefore, there is a significant need to create lightweight and computationally efficient detection mechanisms to permit real-time monitoring without overtaxing the device's resources. A continuing challenge is designing optimally performing models that also balance accuracy and efficiency.

6.2 Data Privacy

Moreover, network traffic data transmitted to central servers from IoT devices can contain sensitive information and represent a significant privacy concern. Centralized intrusion detection systems require raw network traffic data to be aggregated from multiple devices and this greatly increases the risk of permissionless access and/or data exposure. Federated learning can aid in reducing the aforementioned privacy risks by permitting devices to train models using only the model parameters and not the actual data from the devices, which maintains confidentiality regarding the data being transmitted [5].

However, federated learning does still have privacy risk exposures such as information leakages from the model updates and through model inference attacks. Ensuring strong privacy protection while maintaining model performance remains a critical challenge in distributed IoT security frameworks [16].

6.3 Security Issues

IoT networks are highly exposed to cyber attacks such as botnet attacks, distributed denial-of-service (DDoS) attacks, and unauthorized access due to weak security mechanisms. Attackers may take advantage of weak security around IoT devices and use that as a gateway to conduct large coordinated cyberattacks. In addition, federated learning-based systems are also vulnerable to some types of cyberattacks, such as model poisoning attacks where malicious participants modify the global model by submitting inaccurate updates to the model. Model poisoning attacks significantly reduce the intrusion detection system's (IDS) accuracy and reliability [9].

The heterogeneous nature of IoT devices and communication protocols makes implementing uniform security protocols across the entire network challenging. Critical issues related to developing scalable and reliable IoT attack detection systems include ensuring secure communication, protecting the integrity of the model, and accounting for adversarial attacks on the model [3].

7 Future Directions

7.1 Federated Learning

Centralized intrusion detection systems are not well suited for large-scale and dynamic IoT environments due to scalability and privacy limitations. Modern IoT systems require security frameworks that combine privacy preservation with scalability. Federated Learning (FL) enables collaborative model training across distributed devices without sharing raw data, thereby reducing communication overhead and preserving data confidentiality.

Federated learning provides devices on the edge of an IoT network the ability to perform computation on their own datasets, as well as contribute to a global model. This is beneficial to both resource-constrained devices and those that contain sensitive data regarding their users. However, models produced in FL-based systems may exhibit performance issues, since the FL model is trained using data that are stored in a variety of non-IID distributions across many devices. Future research must focus on enhancing scalability and performance, along with improving resilience to adversarial attacks. In addition, using adversarial training methods in conjunction with FL systems will help to create a more robust defense against more complex cyberthreats[1].

7.2 Explainable AI

IoT intrusion detection systems use machine and also deep learning models behave as complex black box models, thus making it difficult to interpret their decision processes. For security-critical applications, transparency and also interpretability of the models is vital to establish trust whilst providing an assurance of reliability.

Explainable artificial intelligence (XAI) gives insight into the predictions made by IoT intrusion detection system (IDS) models. This can be by identifying which features of the input data have contributed to the model's output. Through the application of XAI techniques, security analysts can interpret the behaviour of models, identify weaknesses in the models, and ultimately increase the reliability of the overall IDS system. Furthermore, explainability will help to build trust in automated detection systems, and support improved operational decision making in federated learning environments[9].

7.3 Adaptive Systems

IoT environments are characterised by their high degree of dynamism and heterogeneity, which poses a major cybersecurity challenge. Attackers are continually evolving their attack strategies by exploiting previously unknown vulnerabilities and executing multiple stages of an attack. Consequently, IDS must continue to evolve and become adaptive to the changing nature of the data and emerging attack patterns.

Adaptive learning systems provide ongoing updates to IDS models based upon new incoming data and progressive threats. In a federated learning environment, model training occurs locally at distributed nodes, and these results are then combined through aggregation to provide a single global model, which collectively learns from the local training data whilst utilizing common data across a number of diverse varying network conditions. Incorporating adversarial training and continuous learning strategies can significantly enhance the effectiveness of IoT security systems [1].

8 Conclusion

The Internet of Things (IoT) has experienced unprecedented growth, as many different application types continue to develop. However, this rapid growth of IoT has created serious security concerns for connected devices and their networks. By utilizing an intrusion detection system (IDS), it's possible to identify cyber threats in real-time; however, traditional (centrally-based) IDS solutions are limited in terms of scalability, require excessive amounts of communication, and raise privacy concerns in relation to distributed systems.

Federated learning could be an effective solution for addressing these limitations since it allows for the collaborative training of models while still ensuring privacy of the underlying data and reducing the associated overhead costs required for communication. However, challenges exist in terms of device heterogeneity, non-identical independent data (non-IID) distribution, and vulnerability of adversarial attacks. As such, there are many opportunities to develop trusted and secure methods of integrating federated learning and other advanced technologies (such as blockchain) into distributed IoT systems.

In conclusion, a successful combination of federated learning, edge intelligence, and adaptive security solutions present an ideal framework for developing scalable and privacy-preserving IDSs; therefore, it will be imperative to continue to investigate means to address the issues affecting the development of scalable, privacy-preserving IDSs for the sake of providing the protection necessary for the future variety of IoT systems.

References

[1] Shahla Abdulqader. Cafed-net: Cross-adaptive federated learning with dy-namic adversarial defense for real-time privacy-preserving and threat detection in distributed iot ecosystems. *Journal of Soft Computing and Data Mining*, 6(1):58–68, 2025.

[2] Mohammed Naif Alatawi. Safel-iot: secure adaptive federated learning with explainability for anomaly detection in 6g-enabled smart industry 5.0. *Electronics*, 14(11):2153, 2025.

[3] Nurtay Albanbay, Yerlan Tursynbek, Kalman Graffi, Raissa Uskenbayeva, Zhuldyz Kalpeyeva, Zhastalap Abilkaiyr, and Yerlan Ayapov. Federated learning-based intrusion detection in iot networks: Performance evaluation and data scaling study. *Journal of Sensor and Actuator Networks*, 14(4):78, 2025.

[4] Abdulrahman A Alshdadi, Abdulwahab Ali Almazroi, Nasir Ayub, Miltiadis D Lytras, Eesa Alsolami, Faisal S Alsubaei, and Riad Alharbey. Federated deep learning for scalable and privacy-preserving distributed denial-of-service attack detection in internet of things networks. *Future Internet*, 17(2):88, 2025.

[5] Gutti Chandu, Thumula Karthik, and Balbudhe Parag. Federated learning for distributed iot security: A privacy-preserving approach to intrusion detection. *IEEE Access*, 2025.

[6] Sayan Chatterjee and Manjesh Kumar Hanawal. Federated learning for intrusion detection in iot security: a hybrid ensemble approach. *International Journal of Internet of Things and Cyber-Assurance*, 2(1):62–86, 2022.

[7] Yannis Steve Nsuloun Fotse, Vianney Kengne Tchendji, and Mthulisi Velempini. Federated learning based ddos attacks detection in large scale software-defined network. *IEEE Transactions on Computers*, 74(1):101–115, 2024.

[8] Md Alamgir Hossain and Md Samiul Islam. Towards decentralized cybersecurity: a novel privacy-preserving federated learning approach for botnet attack detection. *Blockchain: Research and Applications*, page 100355, 2025.

[9] Ansam Khraisat, Ammar Alazab, Moutaz Alazab, Areej Obeidat, Sarabjot Singh, and Tony Jan. Federated learning for intrusion detection in iot environments: a privacy-preserving strategy. *Discover Internet of Things*, 5(1):72, 2025.

[10] Jianhua Li, Lingjuan Lyu, Ximeng Liu, Xuyun Zhang, and Xixiang Lyu. Fleam: A federated learning empowered architecture to mitigate ddos in industrial iot. *IEEE Transactions on Industrial Informatics*, 18(6):4059–4068, 2021.

[11] Jin Ho Park, Sangthong Yotxay, Sushil Kumar Singh, and Jong Hyuk Park. Poah-enabled federated learning architecture for ddos attack detection in iot networks. *Hum.-Centric Comput. Inf. Sci*, 14(3):24p, 2024.

[12] Milad Rahmati and Antonino Pagano. Federated learning-driven cybersecurity framework for iot networks with privacy preserving and real-time threat detection capabilities. In *Informatics*, volume 12, page 62. MDPI, 2025.

[13] Christopher Regan, Mohammad Nasajpour, Reza M Parizi, Seyedamin Pouriyeh, Ali Dehghantanha, and Kim-Kwang Raymond Choo. Federated iot attack detection using decentralized edge data. *Machine Learning with Applications*, 8:100263, 2022.

[14] Manuel JCS Reis. Edge-flguard: A federated learning framework for real-time anomaly detection in 5g-enabled iot ecosystems. *Applied Sciences*, 15(12):6452, 2025.

[15] Manuel JCS Reis. Scalable intrusion detection in iot networks via property testing and federated edge ai. *IEEE Access*, 2025.

[16] William Schneble and Geethapriya Thamilarasu. Attack detection using federated learning in medical cyber-physical systems. In *Proc. 28th Int. Conf. Comput. Commun. Netw.(ICCCN)*, volume 29, pages 1–8, 2019.

[17] Mohammed Shalan, Md Rakibul Hasan, Yan Bai, and Juan Li. Enhancing smart home security: blockchain-enabled federated learning with knowledge distillation for intrusion detection. *Smart Cities*, 8(1):35, 2025.